ST MARY'S DSG, KLOOF

GRADE: 10 **10 February 2026**

PHYSICAL SCIENCE CLASS TEST
WAVES

TIME: 42 MINS **TOTAL: 45**
ASSESSOR: T. Kirsten **MODERATORS: K. Bolton and G. Main**

NAME: MEMO

SCIENCE TEACHER: _____

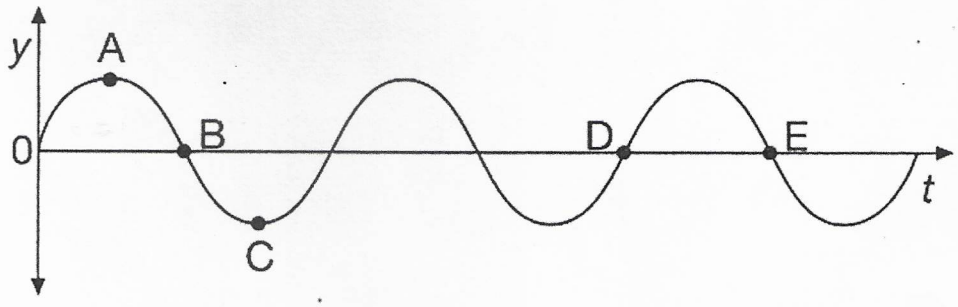
PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 7 pages. A data sheet is provided.
2. Answer on the question paper in the spaces.
3. Write your name and your science teacher's name on your question paper.

QUESTION 1 MULTIPLE CHOICE - Choose the letter that you consider to be most correct and mark with a cross on the grid provided.

1.1	A	B	C	~~D~~
1.2	A	B	~~C~~	D

1.1 The diagram shows a transverse wave. Which of the following pairs of points are **in phase**?



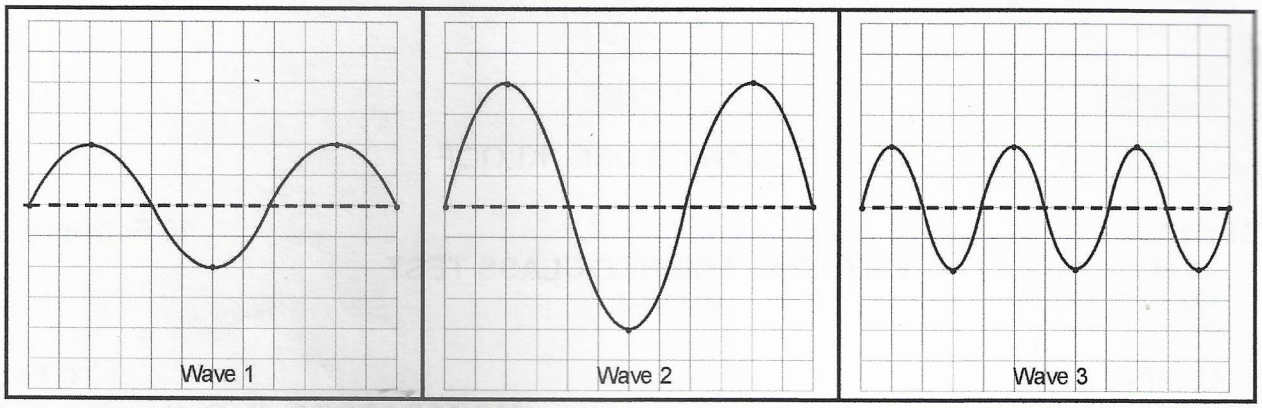
A A and B

B A and C

C B and D

D B and E

1.2 Refer to the three oscillograms shown below:



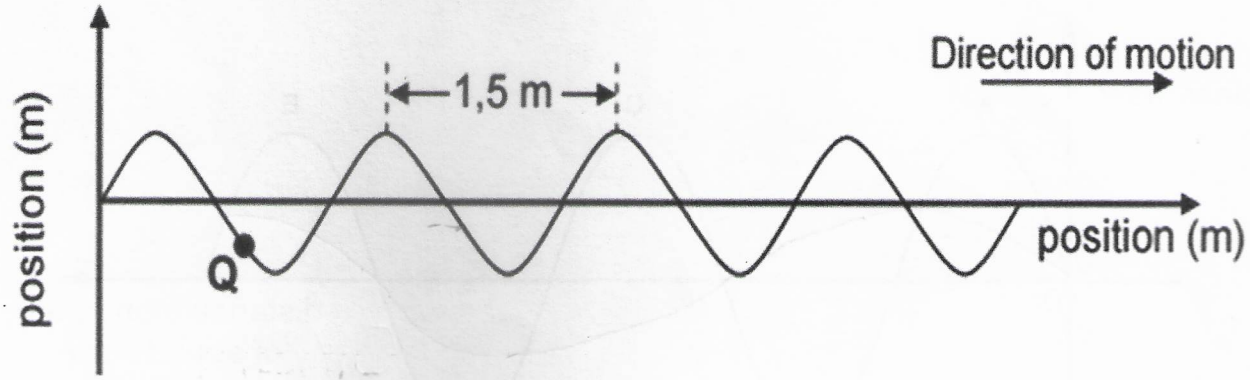
Which of the following is true?

	The wave showing the SOUND OF HIGHEST PITCH is:	**The wave showing the LOUDEST sound is:**
A	Wave 2	Wave 3
B	Wave 3	Wave 1
C	Wave 3	Wave 2
D	Wave 1	Wave 2

2 x 2 = [4]

QUESTION 2

In the sketch below, not drawn to scale, Q represents a cork on the surface of the water in a dam. A person standing on a bridge observes object Q moving up and down. Object Q rises to the top every 5 s.



2.1 In which direction is object Q about to move? (1)

Up ✓

2.2 Define *frequency*. (2)

The number of complete waves ✓ passing a point in 1s ✓

2.3 Calculate the frequency of the wave. (2)

$f = \frac{1}{T} = \frac{1}{5} = 0,2\ Hz$ ✓

2.4 Calculate the speed of the wave. (3)

$v = f\lambda$ ✓

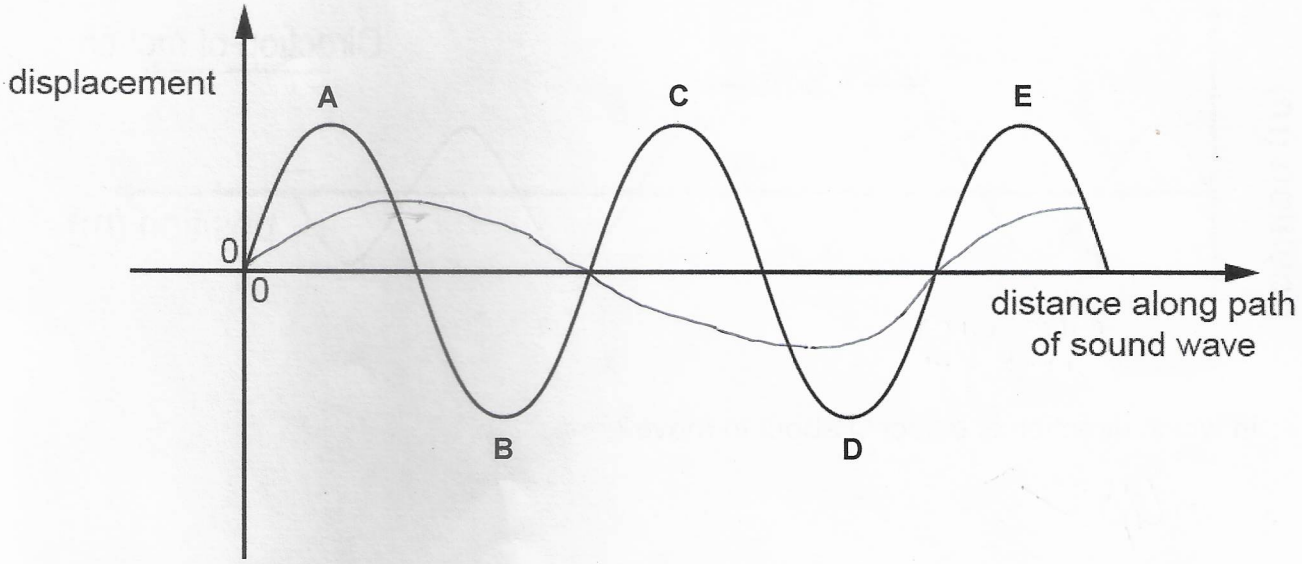
$= 0,2 (coe) \times 1,5$ ✓

$= 0,3\ m \cdot s^{-1}$ ✓

[8]

QUESTION 3

The wave trace below represents how the displacement of air molecules, at an instant in time, varies with distance along the path of a sound wave. The speed of sound in air is 340 $m.s^{-1}$.



3.1 State the type of wave of which a sound wave is an example. (1)

Longitudinal ✓

3.2 What is meant by a *compression* in a sound wave? (1)

Area where particles are close together ✓

3.3 List one letter on the wave trace that would correspond to a compression in the sound wave. (1)

A/C/E ✓

3.4 Define *period* of a wave. (2)

The time taken ✓ for one complete wave to pass a point. ✓

3.5 If the distance moved by the sound wave between **points A and D** is 3,75 m, calculate the period of this sound wave. Write your answer in scientific notation. (5)

$\lambda = \frac{3,75}{1,5} = 2,5m$ ✓

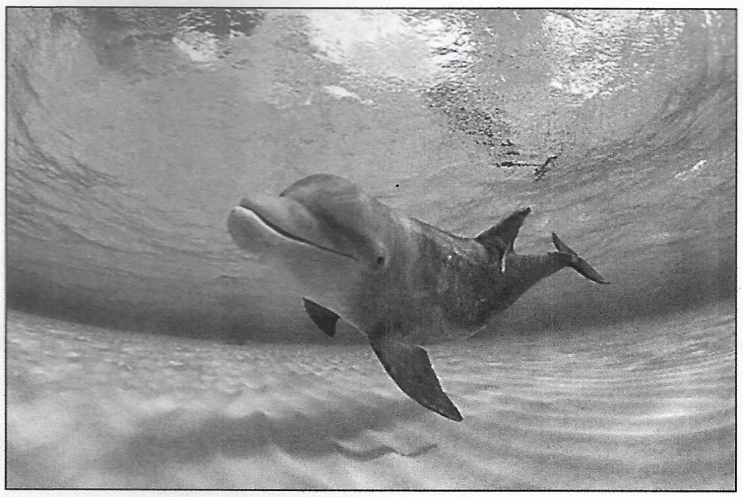
$f = \frac{v}{\lambda} = \frac{340}{2,5} = 136\ Hz$ ✓

$T = \frac{1}{f} = \frac{1}{136}$ ✓(method) $= 0,00735\ s$ ✓

$= 7,35 \times 10^{-3}\ s$ ✓

3.6 On the wave trace on pg 4, draw a corresponding wave which would have ***double the wavelength*** and ***half the amplitude*** of the original wave trace shown. ✓✓ (2)

3.7 A dolphin has a sonar system that emits sounds with a frequency of 21 000 Hz. The dolphin emits a sound which reflects against a shoal of fish, and it returns to the dolphin 0,26 s later. The speed of sound in seawater is 1500 $m.s^{-1}$.



Calculate how far the dolphin is from the shoal of fish. (4)

$\Delta x = v \times t$ ✓

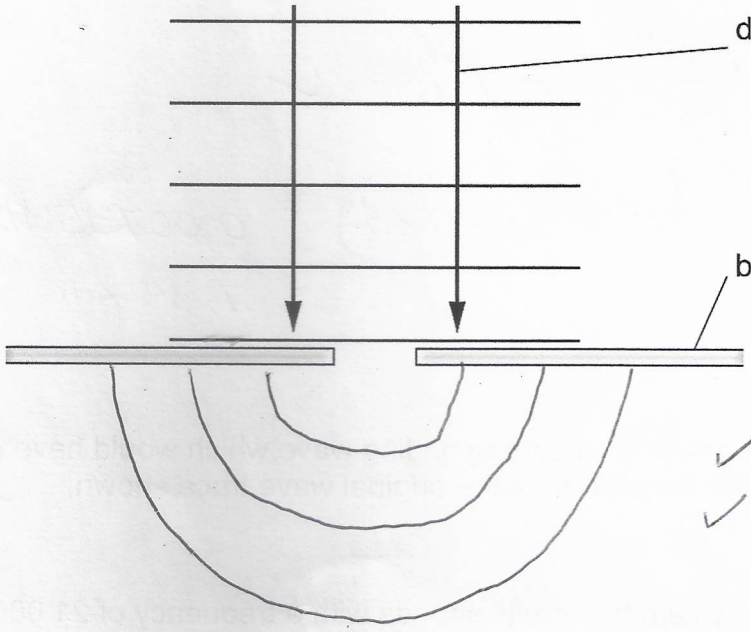
$= 1500$ ✓ $\times \left(\frac{0,26}{2}\right)$ ✓

$= 195m$ ✓

[16]

QUESTION 4

The diagram below shows wavefronts approaching a gap in a barrier in a water tank. The speed of the waves is constant at 0,25 $m.s^{-1}$. The wavelength of the wavefronts is 3,8 cm.



✓ same width
✓ diffraction pattern

4.1 Explain what is meant by *diffraction* of waves. (2)

The spreading out/bending of waves ✓ when passing a barrier or going through a gap ✓

4.2 In the space below the diagram above, sketch the pattern of the wavefronts after the wave has passed through the gap. (2)

4.3 The speed of the waves remains constant, but the **frequency** is then **doubled**. Calculate the **NEW** frequency of the waves. (4)

λ is halved $\therefore \frac{3,8}{2}$ ✓(method) = 1,9cm = 0,019m ✓ (conv)

$f = \frac{v}{\lambda} = \frac{0,25}{0,019}$ ✓(method)

$= 13,16\ Hz$ ✓

OR $f = \frac{v}{0,038}$ ✓ ✓ $= 6,57\ Hz \times 2$ ✓ $= 13,16\ Hz$ ✓

4.4 Describe **how** and **why** the pattern of the wavefronts would change after the wave has passed through the gap. (2)

Shorter λ, ∴ less diffraction ✓ ∴ waves would be more spread out / less bent ✓

[10]

QUESTION 5

Three types of electromagnetic radiations are given in the table below.

Radiation
X-rays
Ultraviolet
Infra-red

5.1 Give one (bold) characteristic that all electromagnetic waves have in common. (1)

Transverse waves / all travel at $3 \times 10^8 m \cdot s^{-1}$ / can be reflected etc / do not require a medium ✓ (any 1)

5.2 Write down the NAME of:

5.2.1 The radiation with the longest wavelength from the given list. (1)

Infra-red ✓

5.2.2 The radiation whose photons have the greatest amount of energy. (1)

X-rays ✓

5.3 Calculate the energy of a photon of light which has a wavelength of 700 nm. (4)

$$E = \frac{hc}{\lambda} \checkmark$$

$$= \frac{6,6 \times 10^{-34} \times 3 \times 10^{8} \checkmark}{7 \times 10^{-7} (conv) \checkmark}$$

$$= 2,83 \times 10^{-19} J \checkmark$$

[7]